# Assignment no 03

**Problem Statement :** The API is working from our developer machine but not from the production server.

**Troubleshooting flow:**

1**.Confirm the issue**:Get API url,HTTP method,status code,wheter all endpoint failure or only one

2**.DNS:**checks whether API hostname resolves to an IPaddress  production server

3.**TCP connectivity**:checks whether the server can connect api port or not,usually it is 443 for HTTPS

4.**TLS/HTTPS:**if connection reaches the host but HTTPS fails,check the certificate validity,hostname,TLS compatibility and server clock

5.**Firewall/Proxy:**check the firewall rules,proxy setting ,VPN ,routing  , production server must be of public IP

6.**Authentication/Authorization**: check whether the credentials are missing/expire/invalid or valid but lack of permission

7.**Endpoint and RateLimit**: Verify URL path,HTTP method,header and requestbody.check the rate limit response on 429

8.**Server logs:** compare the production setting with the developer machine.

## Tools and Commands:

nslookup: Checks DNS resolution.

ping: checks that it is reachable or not

Test-NetConnection:checks the tcp connectivity on windows

Curl: Tests an HTTP/HTTPS request and shows response details

Postman: Replays a request with the correct method, headers, body, and authentication

Application/server logs: Shows timestamps, exceptions, status codes, and request IDs.

## Evidence to request from the client:

**1.**Exact endpoint URL

2.Timestrap when the failure is occurred

3.Full error message,HTTP status code,response body

4.DNS lookup and TCP port test result from production server

5. Relevant application, proxy, gateway, or server logs

6.Confirmation of proxy,firewall,VPN and whether the isuue affects all the endpoint

## Probable cause Decision Tree:

DNS lookup: Likely DNS/hostname/resolver issue.
DNS works, TCP port test fails: Likely network path, firewall, proxy, VPN, allowlist, or service/port issue.
HTTP 401: Authentication issue: missing, invalid, or expired API key/token
HTTP 403: Authorization/access issue: identity may be recognized but lacks permission or is blocked by policy.
HTTP 404 or 405: Wrong endpoint path or unsupported HTTP method.
HTTP 429: Rate limit exceeded; check usage limits and retry instructions.
HTTP 5xx : Possible API/server/gateway issue; inspect logs, service health, and request IDs.

## Sample response to the client

Hello Team,
We understand that the API works from the developer machine but fails from the production server. To investigate, please share the endpoint and HTTP method, the approximate failure time and timezone, the exact error/status code, and any request or correlation ID. Once we receive these details, we can identify which layer is failing and recommend the next step.
Regards,
Support Team